Operating instructions

ALBANY

ASSA ABLOY

RapidProtect RP2000



6410R0175

©2010 ALBANY Door Systems GmbH.

All rights to the door and the related operating instructions are protected. Reproduction of the door system will be followed by legal action.

These operating instructions may not be reprinted or otherwise reproduced without the written consent of ALBANY Door Systems.

The general terms and conditions of ALBANY Door Systems GmbH shall apply.

ALBANY Door Systems reserves the right to make technical changes.

These operating instructions are an integral part of the door. The original operating instructions were produced in German. This is a translation from the German original.

Manufacturer

ASSA ABLOY Entrance Systems AB

Lodjursgatan 10

SE-261 44 Landskrona

Sweden

Internet : www.albanydoors.com

Email: verkauf.de.albany@assaabloy.com

Contents

1 Introduction 5
1.1 Other applicable documents 5
1.2 Declaration of conformity 5
1.3 Amendments and validity 5
1.4 Storage 5
1.5 Target group 5
1.6 Conventions 5
1.6.1 Warnings 5
1.6.2 Operations 6

2 Safety 7
2.1 General safety information 7
2.2 Personal protection 7
2.3 Operator’s duty of care 8
2.4 Intended use 8
2.5 Environmental conditions 8
2.6 Safety features 8
2.6.1 Electrical safety mechanism 8
2.6.2 Additional safety systems 9
2.7 Residual risks 9

3 Product description of the door 10
3.1 Operation and design 10
3.1.1 Slats 10
3.1.2 Side frames 10
3.1.3 Safety devices 10
3.1.4 Manual operation 11
3.1.5 Safety limit switches 11
3.2 Door components 12
3.3 Control system 12
3.4 Drive unit 13
3.5 Type plate (example) 13
3.6 Technical data 14

4 Operation 15
4.1 Safety 15

4.2 Operation 15
4.3 Faults 15
4.3.1 Mechanical faults 16
4.3.2 Electrical faults 16

5 Assembly and dismantling 17
5.1 Inspection of the delivery 17
5.2 Preparation for assembly 17
5.3 Assembly 18
5.4 Dismantling 30

6 Life cycle 31
6.1 Commissioning 31
6.2 Maintenance and cleaning 31
6.2.1 Safety information 31
6.2.2 Inspection plan 31
6.2.3 Cleaning and care 32
6.3 Decommissioning 32
6.4 Transport and storage 33
6.4.1 Transport 33
6.4.2 Storage 34
6.5 Disposal 34

7 Appendix 35
7.1 Technical data 35
7.2 Maximum opening and closing speeds 35
7.3 Diagram of priciples 36
7.4 Fixing plan 37
7.5 Inspection plan 38
7.5.1 Service intervals 39
7.5.2 Annotations to inspection plan 39
7.6 Manufacturer’s certifications 40
7.6.1 Declaration of Incorporation 40
7.6.2 EC Declaration of Conformity 40
7.6.3 Prototype test 40

1 Introduction

These operating instructions are intended for use with doors in the RapidProtect 2000 series with the associated control system. They constitute part of the product and describe how to use the product properly and safely throughout the product lifecycle.

1.1 Other applicable documents

The operating instructions are only valid in combination with the operating instructions for the control system and the inspection book. Both are supplied on delivery.

1.2 Declaration of conformity

The declaration of conformity is included in the inspection book and is supplied with the door.

Modifications that affect the technical data specified in the operating instructions and proper use, i.e. that materially change the machine, will invalidate this manufacturer's declaration of conformity!

1.3 Amendments and validity

The information contained in these instructions constitutes the technical specifications applicable at time of press. Significant changes will be incorporated into a new edition of the user manual. The document and version number of these instructions can be found in the footer.

1.4 Storage

Keep these operating instructions in the storage bag inside the switchbox or, for doors with an MCC control system, in the supplied document box.

1.5 Target group

These operating instructions are intended for trained, qualified and authorised staff. All settings and, in particular, repairs and maintenance, may only be performed by employees of the manufacturer or other trained specialists.

1.6 Conventions

This section explains the conventions used in these instructions for safety information and operations.

1.6.1 Warnings

Warning notices refer to dangers to the life and health of people or to the machine, equipment or the environment.

The notices applied directly to the door must be followed and kept in legible condition.

Warning notices in this document follow the pattern below:

If the danger is an electrical risk, the appropriate pictograms are used:

SIGNAL WORD



Danger

Consequences

⇨ Precautions

SIGNAL WORD



Danger

Consequences

⇨ Precautions

The following warning levels are used:

Signal word	***Hazard level***	***Consequences of non-observance***
DANGER	Immediate grave danger	Death or serious physical injury
WARNING	Potential grave danger	Death or serious physical injury
CAUTION	Potential hazardous situation	Minor physical injuries, material damage

This symbol identifies information that will aid better understanding of how the door works.



Important information

1.6.2 Operations

Aim

1. Step
   → Result of the step (optional)
2. Step…

Result of the operation (optional)

2 Safety

2.1 General safety information

These operating instructions contain important information for commissioning, operating, transporting, storing and maintaining the door safely and properly.

- Read the operating instructions carefully and keep them in a safe place.
- Take particular note of the information in this chapter. Also note the special safety information included in the other chapters.
- Only use the door for the intended purpose. If ownership of the door is passed to a third party, make sure that these operating instructions are also supplied.

2.2 Personal protection

- No special personal protection equipment is required for use of the door. Nevertheless, the door may only be used by trained and authorised staff. Furthermore, the conditions pertaining in the operation where the door is used also apply.
- If the electric drive unit should fail, the door can be opened manually using a hand crank. In this situation, the power supply to the drive unit is cut. The door may partially open or close by itself after the emergency manual lever has been activated. There is the risk of injuries by the movement of the door curtain.
- Do not reach into the side parts of the door or in the area of the top roll when it is in operation.
- Untidiness can cause accidents. Keep the working environment in the immediate vicinity of the door tidy.
- Do not climb the door.
- Whenever work is performed on the door, move the main switch to “Off” or “0” (disconnection from the mains) and secure it against accidental re-activation.
- Stop the door immediately in the event of mechanical or electrical damage, especially if the mains or control cables are damaged.
- Only operate the door at the approved supply voltage and mains frequency.
- Only use accessories or attachments that have been approved by the manufacturer for use with this door.
- Option laser protection: The surface facing to the laser must be kept clean to maintain the protection property.
- Option laser protection: No objects may lie in the closing level, so that a safe sealing is ensured to the ground.

2.3 Operator's duty of care

The door has been designed and manufactured in light of a risk analysis and with careful selection of the standards to be observed and other technical specifications. It is therefore state-of-the-art and guarantees the highest possible degree of safely.

However, this level of safety can only be achieved in practice if all the necessary measures are taken. The operator of the door has a duty of care to plan these measures and check their execution.

Note the following information on safe operation of the door:

- Only employees of the manufacturer or other trained staff may install, commission, inspect, maintain, repair and dismantle the door.
- Only appropriately trained, qualified and authorised staff may operate the door.
- The machine must only be used for its intended purpose.
- The machine must only be operated in perfect operating condition. The safety mechanisms in particular must be checked regularly for operability.
- Always keep the complete operating instructions in legible condition at the location where the door is installed.
- Run regular training for staff on all relevant issues of health and safety, the operating instructions and in particular the safety information contained in the instructions.
- Leave all safety and warning notices applied to the machine and ensure that they are legible.

2.4 Intended use

Unless otherwise specified, our doors have been developed and tested under normal conditions. We are happy to advise you on the choice of suitable options for use under special conditions (e.g. permanent load on one side from temperature, overpressure/vacuum or particular environmental influences etc.).

The door is designed to be a protective barrier and offers the following protection:

- visible barrier for welding installations or machinery centres
- protection from laser beams with laser protection doors
- prevention of movement of persons and access to hazardous areas

2.5 Environmental conditions

Depending on the model, the doors are designed for installation as interior doors and for use under normal environmental conditions.

2.6 Safety features

2.6.1 Electrical safety mechanism

The photocells and control system are subjected to an internal test before the door descends (safety category 2 / performance level c under EN ISO 13849-1 "Safety-related parts of control systems"). If a fault is detected, the power supply to the drive unit is cut.

2.6.2 Additional safety systems

2.6.2.1 Sensors

Depending on the application, it is sensible to add more sensors as safety systems, e.g. for applications with a high level of movement of persons and for doors less than 3 m high that do not have the additional safety mechanism of the user seeing the closing door.

Albany Door Systems offers a wide variety of safety systems for RapidRoll doors including light curtains, motion sensors, induction loops etc. We advise you to ask us for specialist advice on site about your specific application.



A top roll cover is specified for doors less than height DH < 2,5 m.



For smaller doors (DH < 2300mm) which are not mounted on a wall a separate cover for the rear side of the top roll is available as option.

2.7 Residual risks

The door has been designed and manufactured in light of a risk analysis and with careful selection of the standards to be observed and other technical specifications. It is therefore state-of-the-art and guarantees the highest possible degree of safely. However, this level of safety can only be achieved in practice if all the necessary measures are taken. In particular, when the door is operated manually, these persons must have been instructed in the use of the door and the potential danger involved in walking under the closing door must be highlighted.



The door is not designed for constant manual activation.

3 Product description of the door

The following sections explain the operation and design of the RapidProtect 2000. For further technical data refer to the appendix.

3.1 Operation and design

The door RapidProtect 2000 us a vertical opening high speed roll up door for machine protection applications, designed for high operation speed and high number of cycles.

The door curtain is made of aluminium slats connected with reinforced flat belts. The upper ends of these belts on both sides of the door are connected to two special spiral modules which support the first layer of slats. Between each other layer a distance profile is positioned to prevent direct contact with itself and allows to operate the door at high speeds.

The reinforced lifting belt has two basic functions. It serves as the lifting mechanism for the door curtain and it is the connecting point for all of the door curtain slats. In addition the slats are supported by special sliding bands integrated in the side frames.

The lifting power is transferred equally thru the spiral modules to the lifting belts via the drive unit & welded drum tube assembly. This is also supported by the extension spring counter balancing system located within the side frames. With this counterbalancing system a quiet and smooth door operation can be achieved.

The electrical cables for the Contactless Safety Edge are running through a flexible cable chain assembly located in the side frame assemblies.

3.1.1 Slats

The slats of the RapidProtect 2000 are partially double wall anodized aluminium extrusions. Between each slats strong plastic extrusions connect and seal the slats.

Each slat is fixed to the lifting belt individually and can be replaced without requiring lateral clearance space. As option slats with Polycarbonat windows are available.

3.1.2 Side frames

The side frames of RapidProtect 2000 are made from galvanized steel. As option they can be coated in various RAL colours.

In the lintel area both side frames are connected by a U-shaped profile which supports a correct distance between the side frames and can serve as a cable duct.

As option the door can be floor mounted by means of webbed foot plates without needing a separate supporting construction.

3.1.3 Safety devices

All high performance roll doors are equipped with a safety device according to European standards for doors on the bottom slat.

The bottom slat of the RapidProtect 2000 is equipped with a pre-running photocell. Disruption of this photocell leads to a complete stop of the door curtain. After the door curtain stops it will reverse to the open position.

As option RP 2000 can also be equipped with a stationary photocell. Disruption of this photocell will lead to a complete stop of the door curtain, then the door will reverse to the open position.

As an additional option the door can be furnished with an electrical safety edge on the bottom slat. Activation of this edge stops movement of the door curtain, then the door will reverse to the open position. The contact forces with this system are below the allowed values of the norm. The daylight height is reduced by about 90 mm with this system

Before each downward cycle of the door an internal “self testing” of the photocells and the control unit is made automatically. If a fault is detected, disruption of power supply for the control unit occurs immediately.

The extension spring counter balanced system is located in a separate side frame chamber completely surrounded by metal covers.

The doors have a multi independent counter balancing system. Disengagement of the drive unit will not lead to a sudden drop of the door blade. The doors can be partially opened by releasing the brake via the side frame mounted egress lever. The door will partially open by itself and may be opened further by manually pushing the door blade upward.

3.1.4 Manual operation

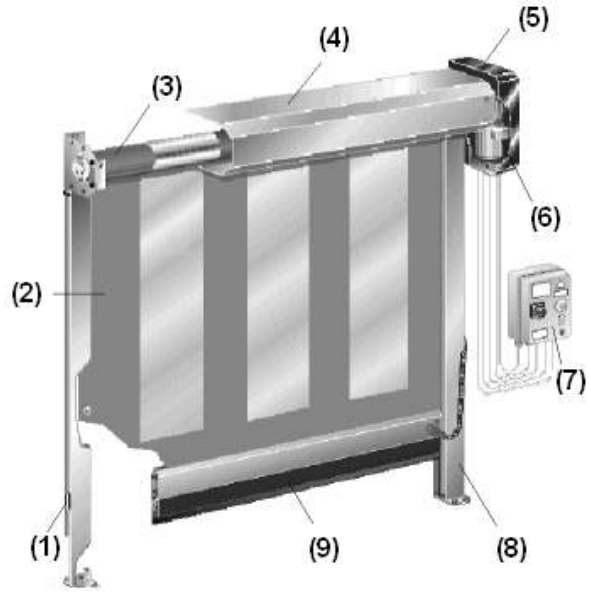
During maintenance work or in the event of a power failure the doors can be partially opened by manually releasing the brake. The door will partially open by itself and may be opened further by manually pushing the door blade upward

3.1.5 Safety limit switches

For integration in automated processes safety limit switches can be integrated in both side frames to allow a safe monitoring of door in closed position independent from the door control system.

3.2 Door components

The RapidRoll 300 door is depicted; details of RapidProtect2000 doors may vary.



1 Safety photocell
2 Door curtain
3 Top roll
4 Optional: roll cover, required for doors DH < 2,5m
5 Optional: motor cover Standard for doors DH<2,5m
6 Drive unit
7 Electrical switchbox
8 Side frames
9 Continuously monitored safety on closing edge

The door line photocell **(1)** monitors the closure area of the door. The door is prevented from closing or stopped if the light beam is broken.

The door curtain **(2)** is made from aluminium slats on doors with rigid curtain.

The top roll **(3)** is based on a special DiscDrive System on doors with rigid curtain, which avoids direct contact of the individual layers of slats when opening the door..

The drive unit **(6)** contains an electric motor, transmission, limit switch and pulse counter.

The electrical switchbox **(7)** contains the MCC control system.

The side frames **(8)** are made of galvanised steel and include the extension spring counter balanced system.

If the safety edge **(9)** is touched, the door immediately returns to the open position. For door model RapidProtect 2000 pre-running photocells are used as standard equipment.

3.3 Control system

RapidProtect 2000 is operated by a frequency converter control system which enables smooth acceleration and deceleration of the doors. See the table in the appendix for the maximum closing speeds of the individual door types.

DANGER

Danger of electrocution

Doors that are not operated with the recommended drive unit or control system can cause electrocution.

⇨ Only operate the door with an original drive unit and a control system from the manufacturer.

3.4 Drive unit

The RapidProtect 2000 is operated with an electric drive unit. The drive unit is fitted at the top of the door, in a position that cannot be reached from the floor, with vibration dampening. The drive unit can be fitted on the left or right as the customer wishes. The temperature of the drive unit is monitored by a thermocouple that cuts the power supply to the drive unit if it overheats.

3.5 Type plate (example)

ALBANY
ASSA ABLOY

ASSA ABLOY Entrance Systems AB
Lodjursgatan 10, SE-261 44 Landskrona
Sweden • Werk 301
Web: www.albanydoors.com
Mail: info.ads.de@albint.com

Kraftbetätigtes Tor:	EN 13241-1
Tortyp:	RP2000
Auftragsnummer:	0030047651
Serien-Nr.:	000000001200000494
Jahr:	20100817
Schaltplan-Nr.:	22651_30047651_040
Steuerung	
Typ:	MCC
Spannung:	400_V_3_Phase
Frequenz:	50 Hz
Max. Leistung:	3 KVA
Schutzklasse:	IP 55
Temp. Bereich:	-20°C... 40°C
Widerstand gegen Windlast:	3
Wasserdichtheit:	0
Luftdurchlässigkeit:	0
Wärmewiderstand:	0

CE Made in Germany
(89/106/EC
2004/108/EC;2006/42/EC)



(the actual data will vary depending on the specific door)

3.6 Technical data

The appendix contains the following technical data:

Technical data	***Contained in***
Door dimensions	Appendix 7.3 general drawing
Dimensions of the fitted door	
Installation space	
Fixing points	Appendix 7.4 Fixing plan

All other technical data relating to the door can be found in the appendix.



Subject to technical changes arising in the course of product maintenance.

4 Operation

This chapter describes the operation of the door irrespective of the control system used. Please read the respective control system manual for a description of control system.

4.1 Safety

- Do not use the door for any purpose other than that intended.
- Take the door out of operation immediately in the event of damage (mechanical or electrical). This applies in particular to damage to the mains or control cables.
- Only use accessories or attachments that have been approved by the manufacturer for use with this door.
- Also note the safety information in chapter 2.

4.2 Operation

The door has an electric drive unit and can be operated both using the button on the door control system and via additional, external impulse generators and safety mechanisms.

Generally the Open, Stop and Close buttons on the door control system should be used during setup, maintenance and repair of the door.

In normal operation, the door is controlled via external signal sources such as rope pull switches, induction loops, radar or radio sensors or palm buttons connected to the door control system via potential-free contacts.

Machine protection doors can be operated via a higher-level control system.



The choice of impulse generator is part of the design and setup of the door and control system and may only be performed by trained specialists.

4.3 Faults



DANGER



Danger of personal injury and material damage

Improper fault clearance can lead to damage to the product and serious injury to staff.

⇨ Faults should only be cleared by appropriately qualified staff.

⇨ Switch off the main switch to secure the door against accidental re-activation.

⇨ Lock the main switch to secure it against re-activation or ensure safety shutdown of the machine by a second person.

⇨ Secure the area of activity around the door.

If the fault cannot be cleared, contact the supplier's or manufacturer's customer service.

4.3.1 Mechanical faults

Fault	*Possible cause*	*Necessary actions*
Door opens or closes by itself	Brake on motor is defective	Replace brake
	Brake incorrectly set	Set brake
	Balancing system not dimensioned correctly	Call customer service
Malfunction of manual opening	Balancing system defective	Replace balancing system
	Bowden lever disengaged or set too loosely	Fasten bowden lever and tension properly
	Brake of motor is locked	Call customer service
Motor not working	No power supply	Connect the door to the mains
	Electrical connections are defective	Check connections
	Fuse has blown	Replace fuse
Curtain rolls up crookedly	Door has been installed incorrectly	Call customer service
	An object has caught in the rolled curtain	Close the door and check for extraneous objects

4.3.2 Electrical faults

Fault	*Possible cause*	*Necessary actions*
Door not working	No mains power	Check mains power supply/fuse
	Control system fuse blown	Check control system and replace fuse
	Main switch off	Turn on control system (main switch)
	Emergency stop circuit broken	Check emergency stop circuit. Bridge unnecessary stop inputs as per wiring diagram.
Door does not close	Up pulse is permanently on	Check impulse generator
	Safety photocell blocked	Check safety mechanism
Door only closes on next pulse	Alternating switch set	Check set control function

5 Assembly and dismantling

DANGER

Risk of injury

Improper handling of the packaged door can lead to injury.

⇨ Place palette on an even surface.

⇨ Do not remove palette before starting assembly.

5.1 Inspection of the delivery

The door is largely pre-assembled on delivery. The side frames, roll cover, motor cover (if supplied), fixing materials and control system are secured to a transport frame (wooden palette).

- Check door for damage incurred during transit
- Check that the delivery is complete by comparing it with the order documentation.

5.2 Preparation for assembly

DANGER

Risk of injury

Incorrect assembly of the door can lead to injury.

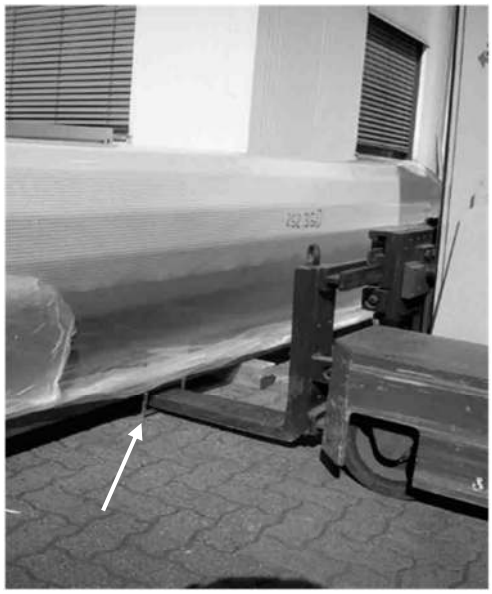
⇨ The door may only be assembled by trained staff.

⇨ The door may only be assembled by the manufacturer or an installation firm approved by the manufacturer.

Make the following preparations before assembling the door:

- Inspect the installation site for the necessary fixing points and assembly requirements: assemble a substructure or other suitable fixing options if necessary. Screw the upper fixing points to a steel substructure or concrete element. The fixing plan can be found in 7.5.
- Measure clearance dimensions (width and height) of the door opening and compare with the order.
- Cordon off the installation site against people and vehicle movements.
- Remove shrink-wrap from the packaged door.

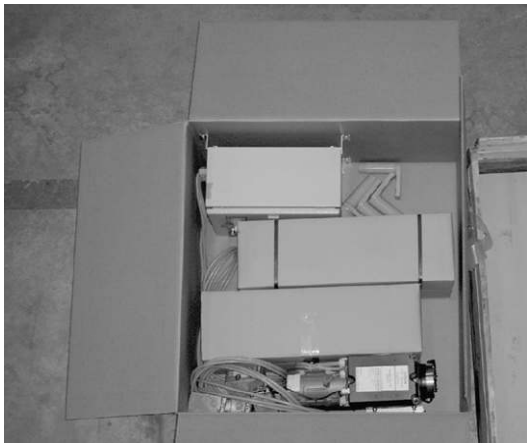
5.3 Assembly



1. The barrel is placed on a pallet which works as a tool for transport and installation, too. The transport-ring is to be used by the forklift driver as shown above to avoid a toppling of the complete top section while transport.



2. The side frames including the springs and the cover plates are located on a second pallet.



4. Type of installation

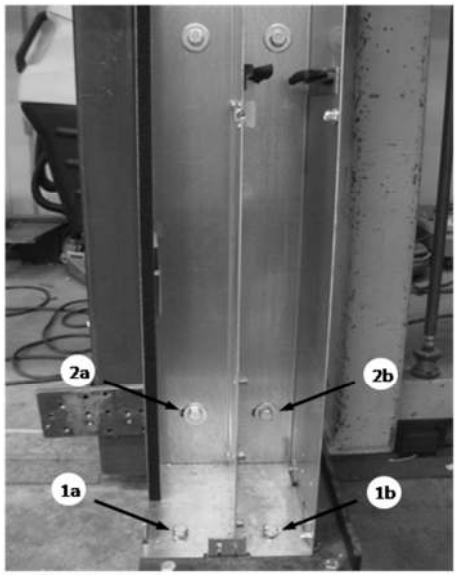
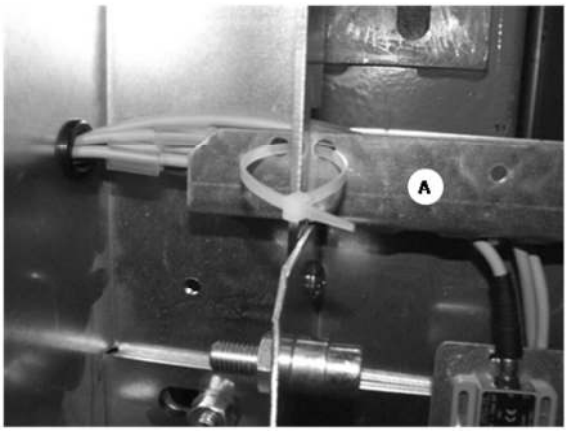
4.1. Wall mounting (Steel construction or equivalent is required)

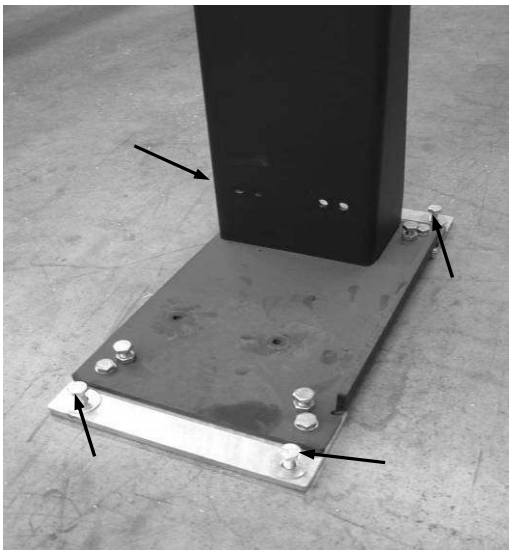
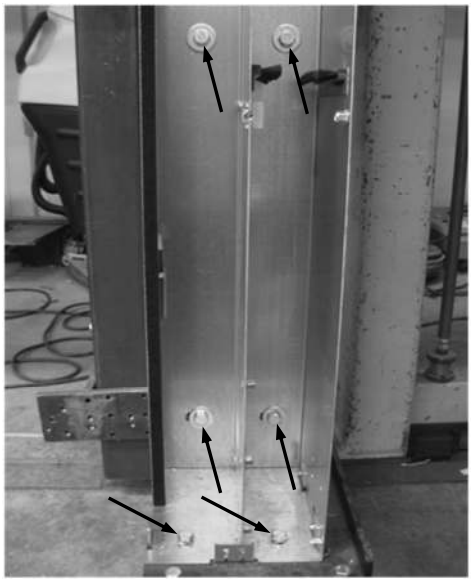
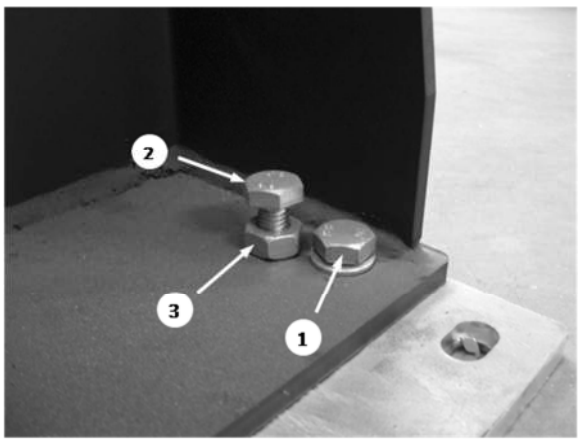
4.2. Floor mounting with webbed foot plates

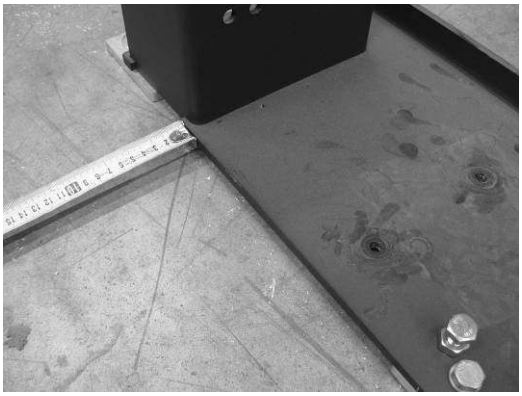
If the floor declines the first side frame has to be arranged and fixed on the side of the higher level.

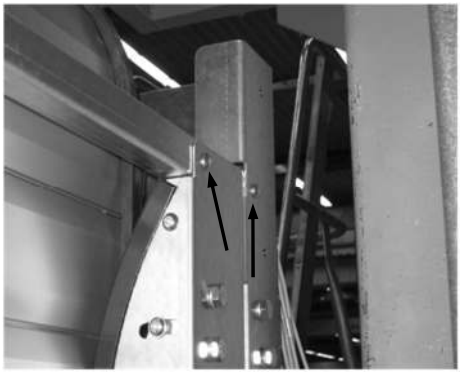
(For fixing points please refer to fixing plan).

3. Control, drive and material for installation is housed in a common cardboard box.

	
5.1 In case of wall mounting the first side frame must be placed exactly perpendicular.	**5.2** Fix the first side frame on the wall within the upper area (min. one screw D ≥ 10 mm per chamber).
	
5.3 Fix the side frame within the lower area with two screws in minimum either strictly on the floor (1) or alternatively on the wall (2).	**5.4** Erect the second side frame and fix it loosely on the wall and floor. Fit the lintel profile (A) into the openings of the head plate. Secure it on both sides by a cable binder.

5.5 Dimension B has to be measured between the inner sides of both side frames. Adjust the side frame at the bottom exactly to daylight width. Fix the side frame completely.	
	6.1 For floor mounting installation fit the webbed foot plates with four screws on the floor (D ≥10 mm).
	
6.2 Fix the side frame with six screws on the webbed foot plate.	**6.3** Loosen the fixing screws (1) and adjust the side frame exactly perpendicular by means of the adjustment screws (2). Fix this setting by tightening the nuts (3) and finally tighten the fixing screws (1).



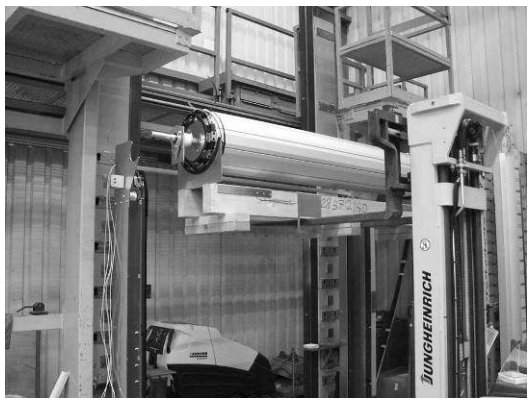


Not required for doors with additional cover at the rear side (see picture 41).

6.4 Position the second webbed foot plate exactly at daylight width and fix it there (6.1). Fix the second side frame on the food plate (6.2) and position it properly (6.3). The light opening width is given by the distance between the two foot plates.

6.5 Insert the lintel profile and fix it with two screws per side.

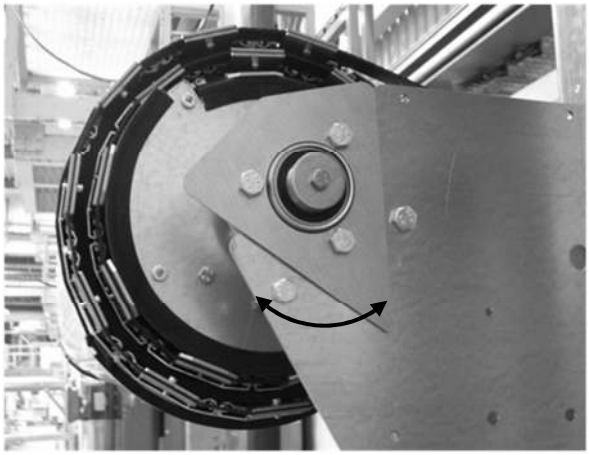


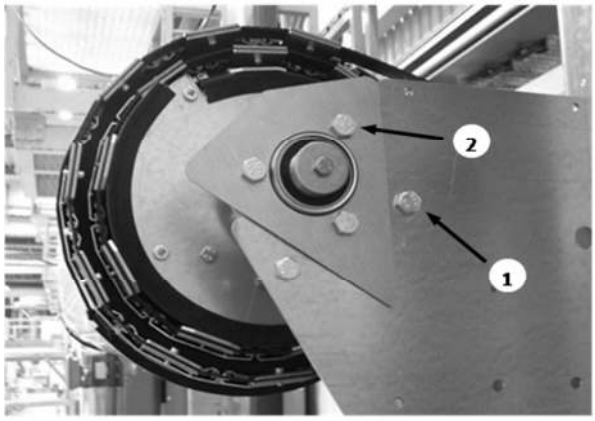


7. Guide the cables of the door line photocell to the drive side. Use the lintel profile as cable duct.

8. Lift up the barrel with assistance of a crane, forklift or equivalent device and lay down the complete barrel on the side frames as shown above.

While take down the pallet again the barrel can start to run.

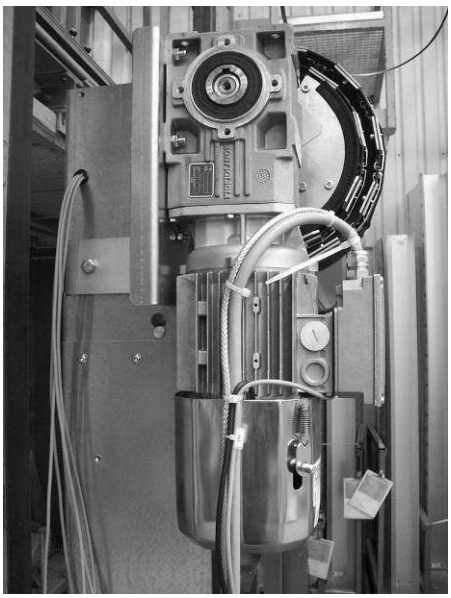


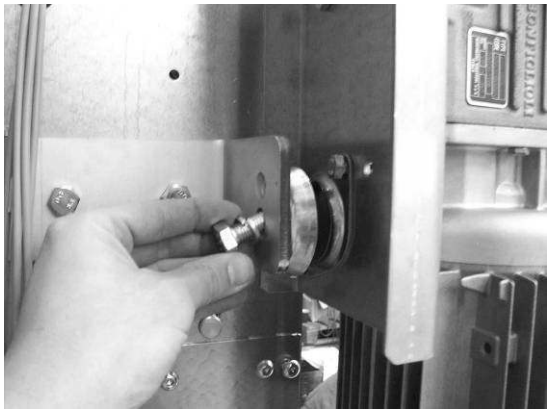


At doors with top roll cover at first install the side cover (See picture 37)

9. The arrow of the mounting plate has to be arranged in direction to the whole, as shown.

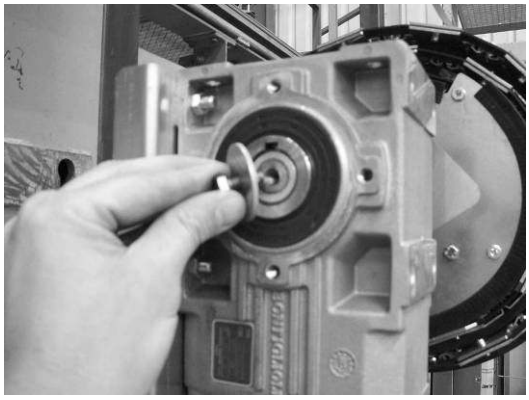
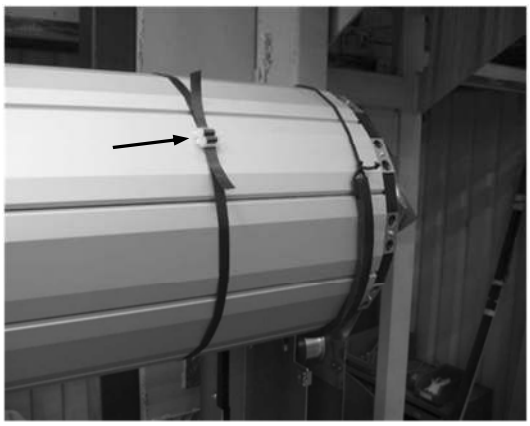
10. Screw on the mounting plate with the side frame (1). Afterwards tighten the three screws of the flange-bearing (2).

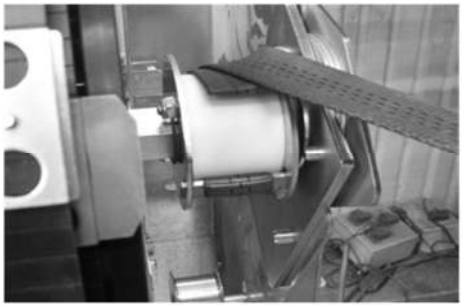


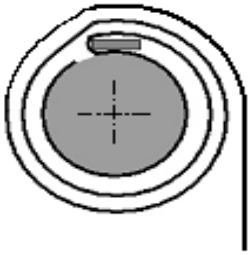


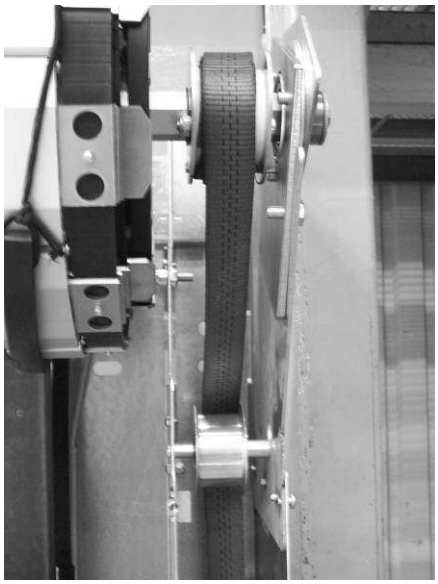
11. Lift the drive and the control together and slide the gear drive (picture 34) onto the shaft of the barrel. Turn the shaft with use of an Allen-key (6 mm) into the correct position.

12. Fix the torque arm on the damper of the drive unit.

	
13. Fix the drive position on the shaft of the barrel.	**14.** For assembly of MCC control system move the fixing bracket over the screws and pull them tight.
	
15. Now turn the top roll with a wrench SW 6 into the correct position (see picture 33) and then remove the PVC-belts which hold the barrel in the transport condition. Correct position: bottom lamella facing to the bottom.	**16.** The fixation of the belts of the counterbalance system on the barrel is given via a strip of metal. Please wind the belts contrary to the direction of rolling up of the curtain.



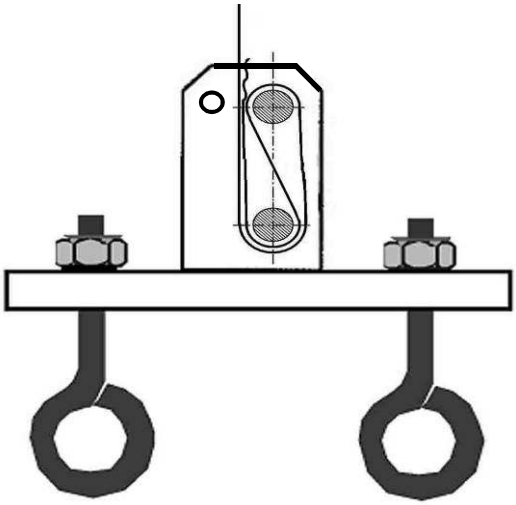




17. Place two dead windings on the barrel.

18. The belts have to be guided perpendicular and free of twist.

The belts must be lead through the deflection pulleys.

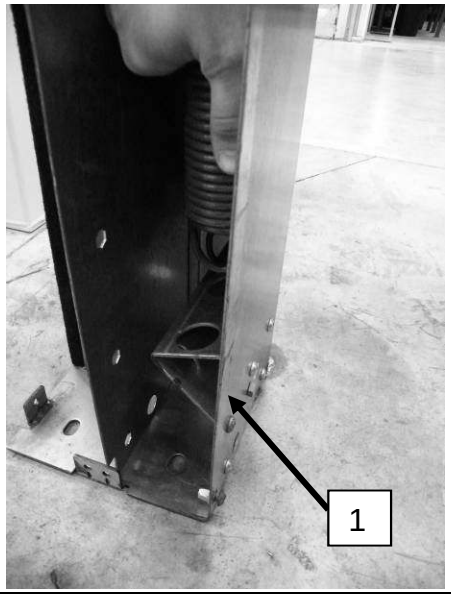
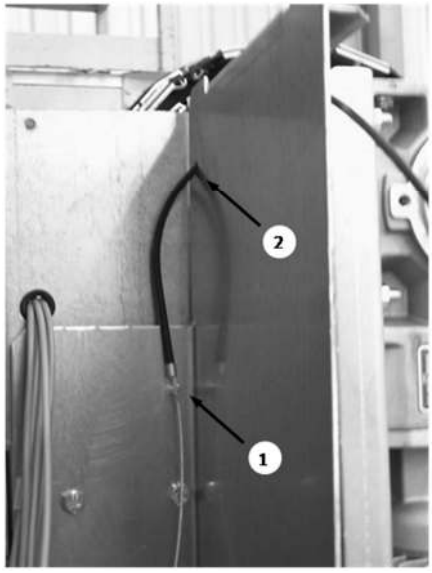


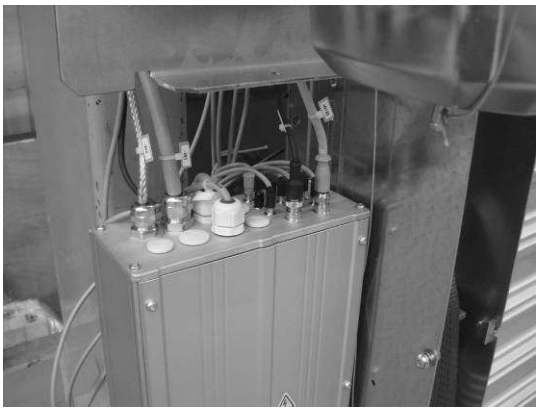


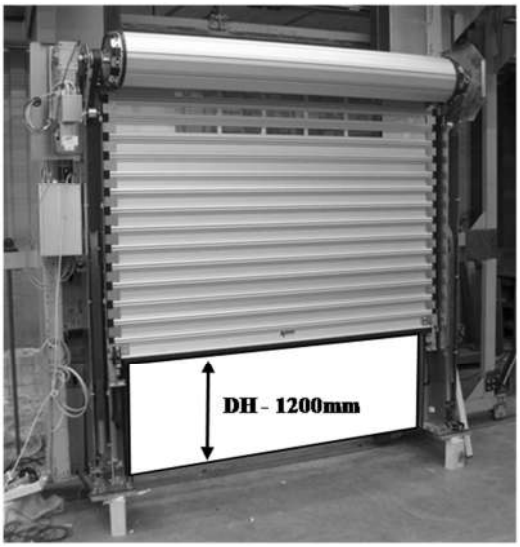
19. The sketch shows the fixation of the belts onto the upper spring acceptance of the counterweight system.

20. Connect the belts with the upper spring acceptances.

The pre-tension of the springs with opened door: 150 mm (= distance between lower eye and lower spring acceptance)

	
21. Hang in the springs in the upper acceptances first before you use the lower acceptances. Connect earth connection from building (min. 6 mm^2) (1)	**22.** Fix the bracket for the motor cover (2 pcs. M8 x 25).
	
23. Pull the steel rope of the bowdencable through the drills of the support (1). Guide the bowdencable through the bracket for the motor cover (2).	**24.** The other end of the cable must be able to affect the pin (3) of the brake as shown above. Then fix is to the cover of the blower (4).

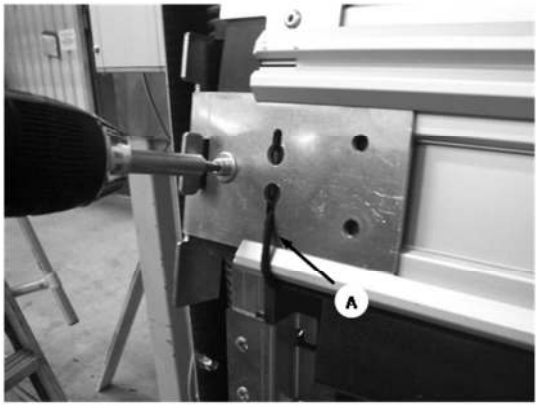


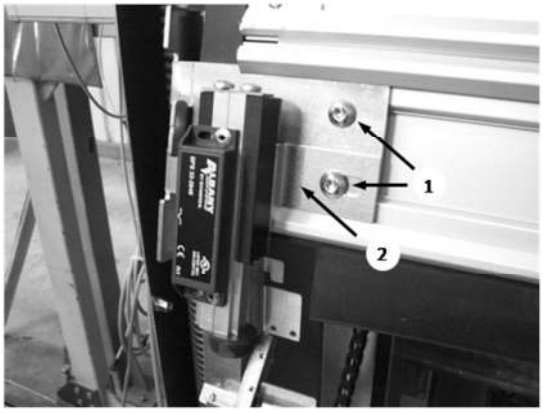


25. Connect the cables to the control system.

⚠ The markings at the plug board must correspond with the ones of the cables.

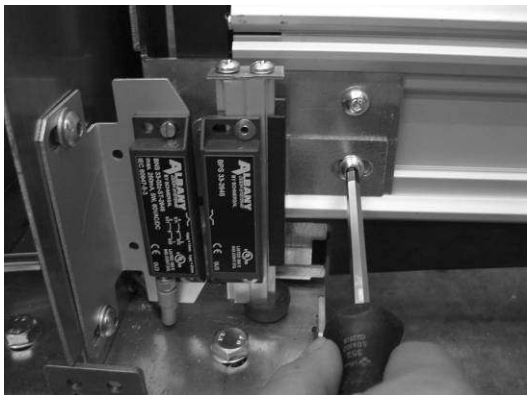
26. Close the door down to approx. 1200 mm above the floor.

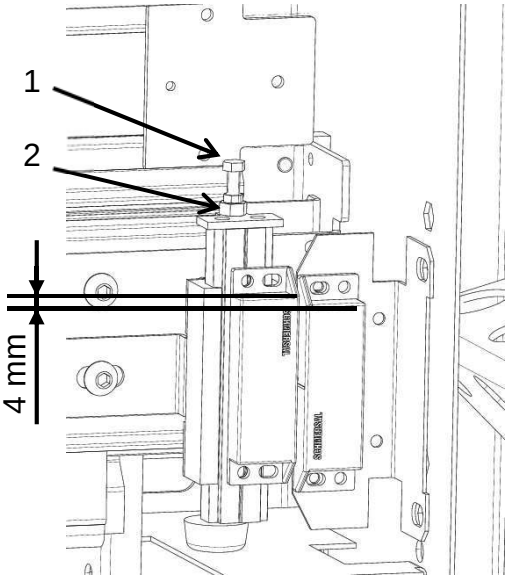




27. Screw the adapter of the bottom lamella to the lifting belt. Mount the rubber rope (A) in the lower key-hole.

28. Fix the adapter onto the bottom lamella (1). Simultaneously fix the support of the optional limit switch.



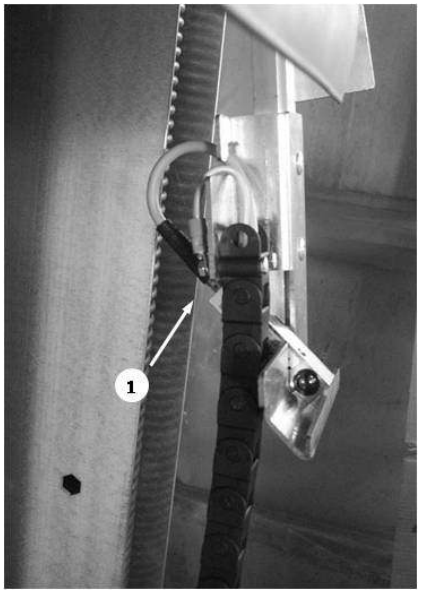


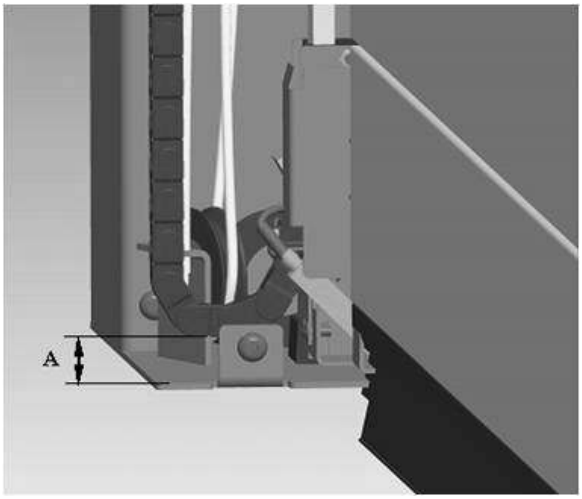
29. Close the door and adjust the position of solenoid and sensor of the limit switch. Both have to be in line and placed in a distance between min. 1 and max. 4 mm.

30. Option laser protection:

Turn adjusting screw (1) as fas as the magnet is 4 mm higher than the switch.

Fix adjusting screw with counternut (2).

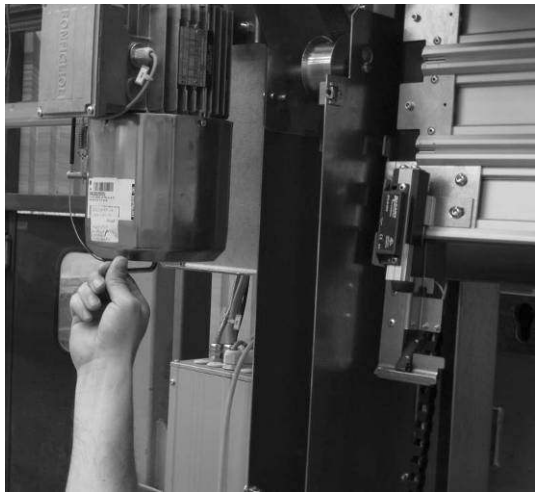
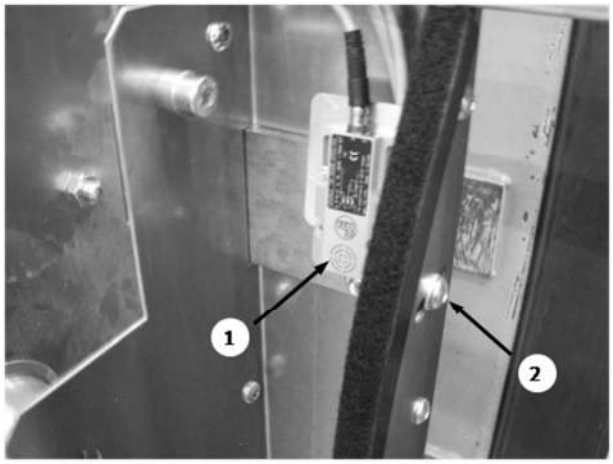


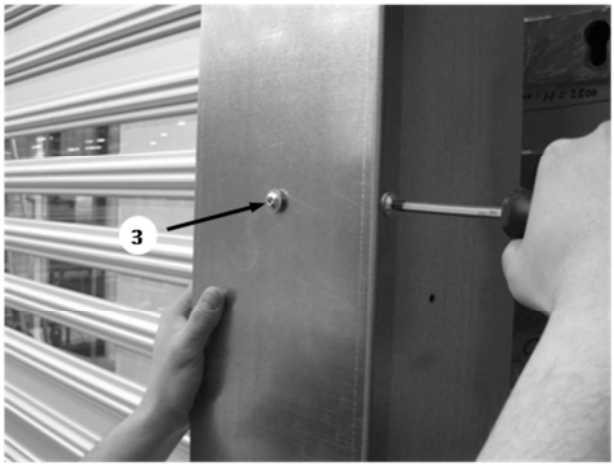
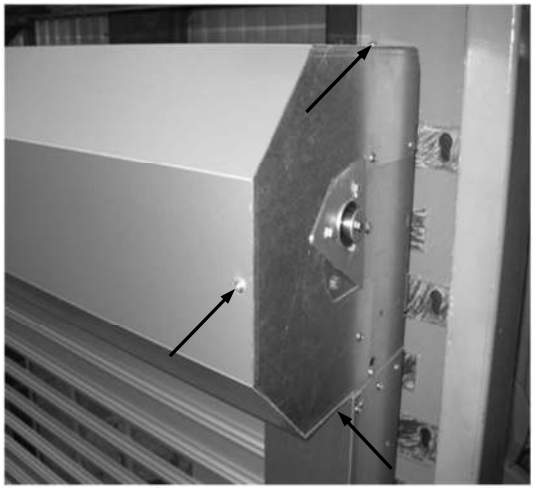


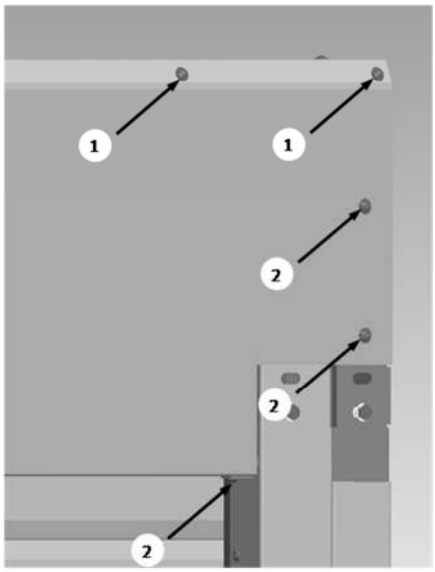
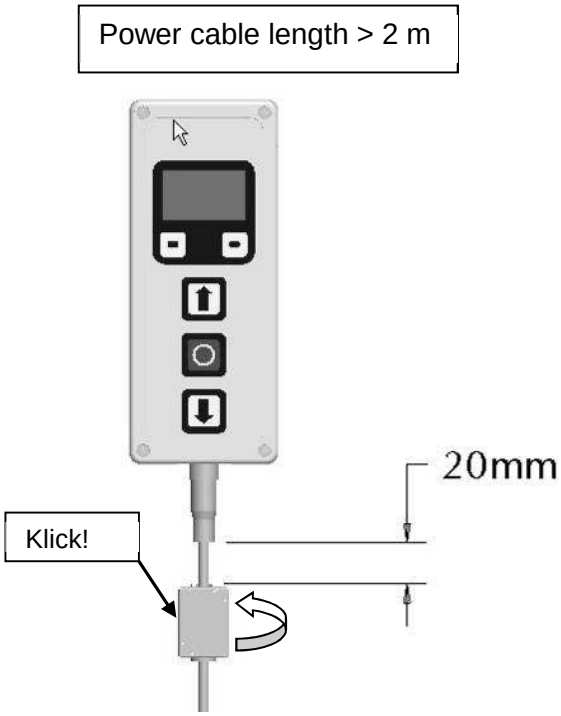
31. Screw together the loose end of the power chain with the bracket of the prerunning photocell and connect the cable.

Pay attention to a good contact of the earth connection (1). Guide the cable in small radius.

32. During assembly of the energy chains ensure that the respective energy chain does not touch the floor in case of closed door (measure of A: approx. 10 mm).

	
33. The door has to be opened up to H minus 300 mm.	**34.** Open the door completely by Allen-key width 6 mm) or alternatively by hand (release brake by pulling the bowdencable).
	
35. The position of the inductive proximity-switch (1) is adjustable via a screw (2) and the corresponding elongated whole.	**36.** Close the door approx. 70 % and mount the side frame covers. Screw them loosely to the front of the spring compartment.

	
37. Push the outer cover plates onto the curtain and press and fasten the screws at the side. Then fasten the screws at the front side (3).	**38.** Fix the lateral covers on both sides of the top roll cover.
	
39. Position the top roll cover and screw it to the lateral covers.	**40.** Mount the motor cover and screw it to the fixation bracket.

	
41. Option: Mount the rear cover for the barrel. Fix it to the top roll cover (1) and the side frames (2).	42.
The installation is finished once all safeties have been tested properly and the successful completion of the test runs.	

5.4 Dismantling

Simply perform steps in the assembly instructions in reverse order to dismantle the door.

To dismantle the barrel a suitable pallet have be ordered in advance.

The material of packaging can be given back to the manufacturer. Spoiled door-components must be wasted by the customer environment-friendly and in harmony with the valid law.

Defect electronic equipment belongs to special waste and because of this specific laws have to be respected in case of disposal.

6 Life cycle

6.1 Commissioning

Please read the information in the appropriate control system manual.

6.2 Maintenance and cleaning

Please contact the supplier or manufacturer for maintenance / fault clearance / repair.

ALBANY Door Systems - Service

Am Mondschein 25

D - 59557 Lippstadt

Germany

Tel.: +49 (0)2941 766 - 0

Fax.: +49(0)2941 766 – 294

Internet : www.albanydoors.com

Email: service.de.albany@assaabloy.com

6.2.1 Safety information



Only the manufacturer or trained staff may maintain, inspect or service the door.

- Disconnect the power supply using the lockable main switch in the switchbox prior to any maintenance and cleaning work.
- Use a testing device to make sure that there is no power to the switchbox.
- Make sure that all scaffolding, ladders etc. comply with the applicable safety regulations.
- Cordon off the work area. When the work is finished, remove any leaked lubricants and tools or other materials from the floor.

6.2.2 Inspection plan

Note all maintenance, repair and modification work in the inspection list that is included in the inspection book for every door to ensure that any changes can be traced.

Document the requisite services, inspections and implementation notes in the inspection plan that can be found in the inspection book.

6.2.3 Cleaning and care

6.2.3.1 Curtain

CAUTION

Risk of material damage

Improper cleaning of the curtain may damage the surface

- ⇨ Do not use glass cleaners for windows as they contain aggressive substances.
- ⇨ Never use abrasive cleaners, scrapers, razor blades, filling knives etc.
- ⇨ Clean with warm water, a low dose of a gentle plastic cleaner and a clean, soft and non-fluffy cloth.

DANGER

Risk of personal injury and material damage

Pollution of the curtain can cause loss of the protection property for laser protection doors.

- ⇨ The surface facing to the laser must be kept clean.

6.2.3.2 Door frame

CAUTION

Risk of material damage

Improper cleaning of the curtain may damage the surface

- ⇨ Do not use pressure cleaners.
- ⇨ Remove dust with a soft cloth.
- ⇨ Remove heavier soiling with water and an ordinary liquid cleaner.
- ⇨ Metal surfaces that are soiled with grease or oil should be cleaned with an agent that contains a solvent.

6.2.3.3 Surrounding area

After maintenance work in particular, the floor in the immediate vicinity of the side frames may be soiled. Remove this with care.

6.3 Decommissioning

- Follow the assembly instructions during dismantling.
- Disconnect the power from the door before dismantling. In particular, make sure that all the springs are de-tensioned.
- Check parts that are subject to wear and tear and replace if necessary before reassembly.
- Check connectors, do not kink cables.
- Store all the components in a clean, dry place.

6.4 Transport and storage

6.4.1 Transport

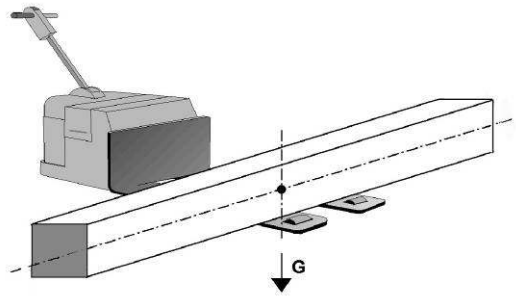
The individual door components may only be removed from the transport palette at the installation site.

DANGER

Risk of injury

Improper transport can lead to the load slipping and falling.

- ⇨ Follow the public safety regulations during transport to the installation site.
- ⇨ Secure the door against tipping and slipping.
- ⇨ Lift the door at its centre of gravity to prevent it slipping sideways.



6.4.1.1 Dimensions and weights

Dimensions of the transport palette and door

	Transport palette [mm]		***Drive unit and control system palette [mm x mm]***
Door	***Length***	***Width***	***Length x Width***
RP 2000 (top roll)	W + 900	750	-
RP 2000 Palette II (Side frames)	H + 400	750	-

(W = Width of door opening) (H = Height of door opening)

Weight of door with palette [Kg] (intermediate values can be interpolated).

W x H [m]	***2 x 2***	***2 x 2,5***	***3 x 3***			
RP 2000 Top roll			250			
RP 2000 Side frames			270			

6.4.2 Storage

The partially pre-assembled door is delivered on a transport/assembly palette and is shrink-wrapped. If the packaging is undamaged, the door can be stored outside for a few days.

6.5 Disposal

The door packaging can be returned to the manufacturer. Defective door components must be disposed of in an environmentally friendly way by the customer.

Defective electronic components must be disposed of as special waste.

7 Appendix

The table below contains the technical data for the RapidProtect 2000 series doors. You can also find a brief summary of the maximum opening and closing speeds in section 7.2

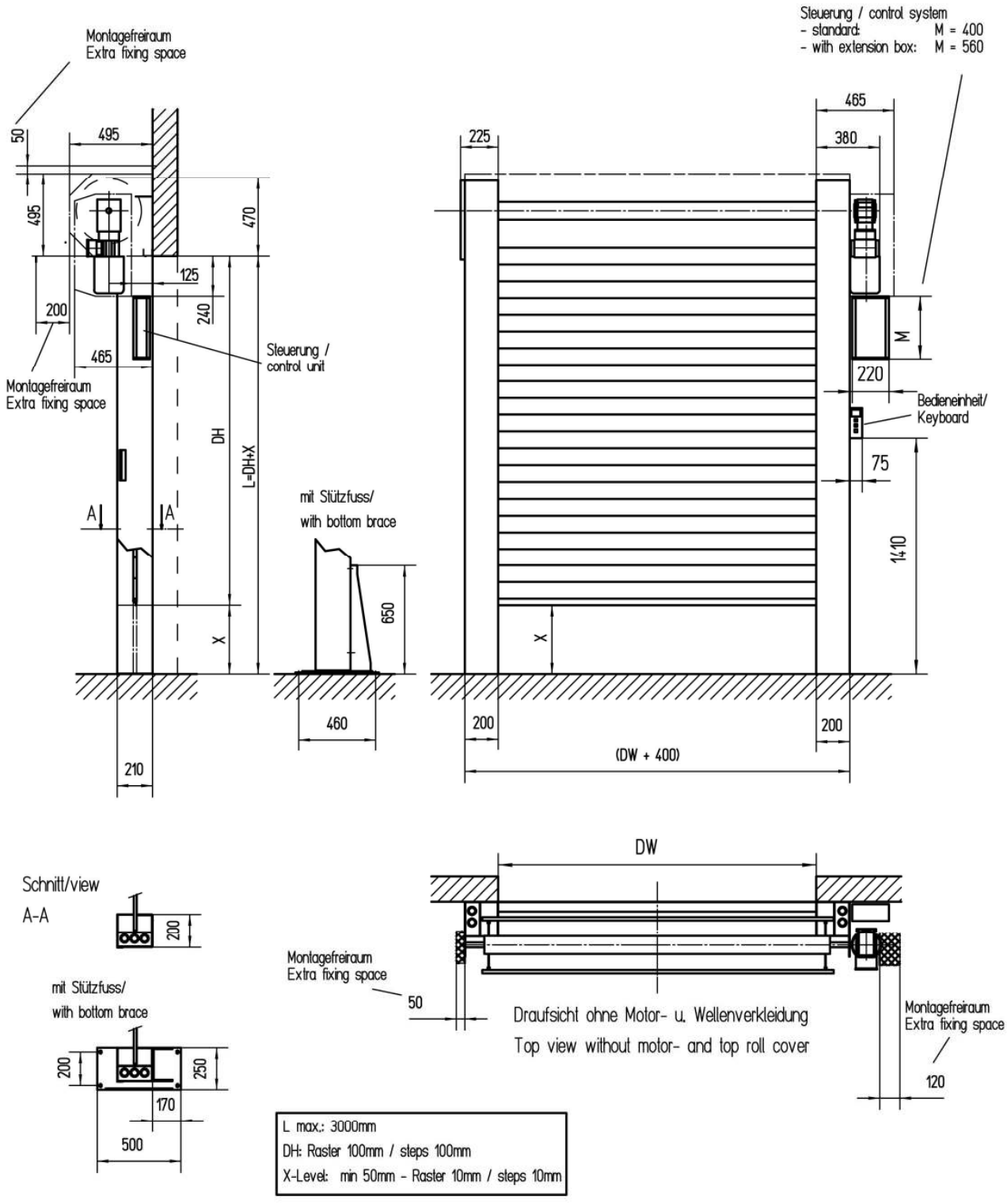
7.1 Technical data

	RapidProtect 2000
Door principle:	Roller door
Direction of movement:	Vertical
Application:	High speed machine protection door
Available sizes:	
W min:	600 mm
W max:	3500 mm
H min:	1000 mm
H max:	3000 mm
max. opening speed:	2,0 m/s
max. closing speed:	ca. 1,2 m/s
Installation site:	Inside a structure
Drive unit:	Electrical
MCC mains connection:	3/(N)/PE 380-480V, 50/60Hz
MCC mains connection with upstream transformer:	3/(N)/PE 220V, 230V, 500V 50/60Hz
Motor power:	1,1 kW
Control system:	MCC
Control supply:	24 V DC
Operating temperature range:	+10℃ - +45℃
Motor protection rating:	IP55
Side frame material:	Galvanised steel
Curtain:	Aluminium, anodised
Lifting belt:	Flat belt with integrated steel ropes
Noise level:	70 dB A
Wind resistance:	Class 3, 700 N/m², (EN 12424)

7.2 Maximum opening and closing speeds

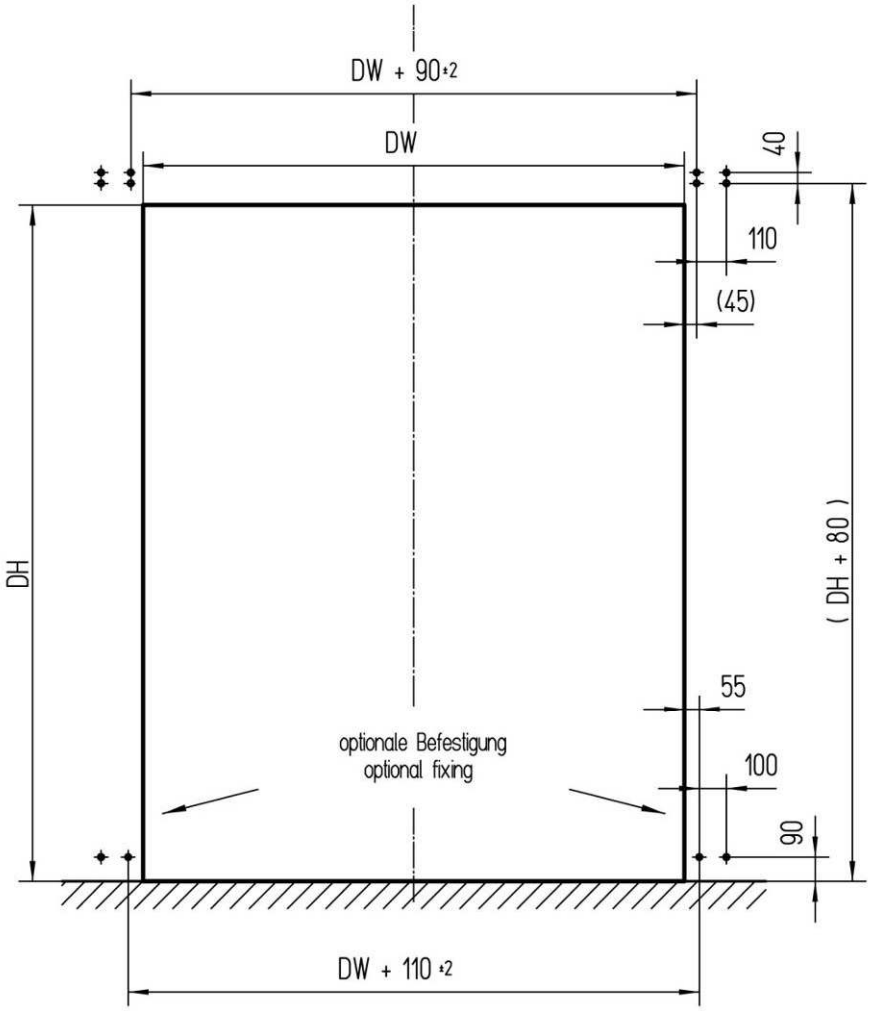
	MCC		***ACS 50***	
m/s (max)	Opening	Closing	Opening	Closing
RP 2000	2,3	1,2	-	-

7.3 Diagram of priciples

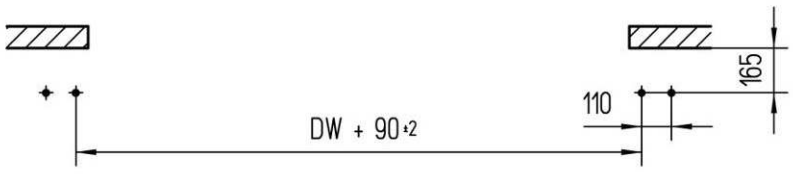


Zg.-Nr.8300R0031/d
Stand: 2012-02-21

7.4 Fixing plan



Befestigungsschrauben M12 oder vergleichbare
fixing screws M12 or comparables



Zg.-Nr.: 8300R0032 Rev.a
Stand: 2009-03-09

7.5 Inspection plan

	Installation fixings	
	Before checking the following mentioned components the cover in front of the spring housing and eventually the existing top roll and motor cover must be dismantled.	2
	Drive unit	
X	Check brake condition and function of brake release lever	2
	Check retainer of the torque support and buffer	2
	Check setting, function and ease of operation of the bowdencable of brake release	2
	Check self-opening / manual opening	2
	Drive shaft	
	Check flange bearing retainers of drive shaft	2
	Check retainer of spiral disk on the shaft	2
	Check condition of sound isolation profiles at the spiral disks	2
	Check flat belts for wear and damage; check connection to spiral disks	2
	Exchange of flat belts	4
	Door curtain	
	Check surface of slats for grinding signs and damages	2
	Check exact position of slat sealing	2
	Check fixing of slats on pulling device	2
	Check solid position of slat adapter	2
	Guiding rails	
	Check for wear & tear on guiding rails, especially in the area of funnel	2
	Replace sliding band, especially in the area of funnel	2
	Side frames	
	Check fixing and condition of the profiles as well as fixing at the wall	2
	Check condition and fixing of trailing chain for cable	2
	Check condition and installation of springs and spring fixing	2
X	Clean optic of photocells	6
	Counter balancing system	
	Exchange springs and belts	3
	Check condition and fixation of belts	2
	Control box and additional components (Ex. actuators)	
	Check completeness of wiring diagram	5
	Check main switch/CEE-plug and control box lock	5
	Check fixing	5
	Electrical components	
	View-check for mechanical damages	5
	Check potential balance	2

	Check function of photocells	1
	Check function and position of safety limit switches	2
X	Clean optic of photocells	6
X	Check the function of the pre-running photocell	2
	Check distance of upper reference switch to cam switch	2
	Mechanical functions	
	Check function and condition of torque support during operation	2
	Review possibility for manual emergency opening	2
	Check winding performance of door leaf the ropes and belts	2
	Movement of the door	
	Check opening and closing position of bottom profile	2

X see following text for annotations

7.5.1 Service intervals

1	half-yearly or 50.000 LW
2	annually or 100.000 cycles
3	500.000 cycles
4	1.000.000 cycles
5	annually
6	clean if necessary

7.5.2 Annotations to inspection plan

7.5.2.1 Brake

If the gap of the brake cannot be adjusted to a distance of 0,2 mm, the brake must be changed.

7.5.2.2 Safety edge / pre-running photocell

Let the bottom profile come down on an object. The door must return to final door position top after touching the object. With prerunning photocell hold the flat hand vertically below the bottom profile in about 1 m height. The door has to reverse in final door position without touching or only slight touching of the rubber profile.

7.5.2.3 Function of photocell

Both LED of photocell must be switched on.

Transmitter green LED, receiver red LED

The red LED of the receiver turns off after interruption of the photocell. The door must go up to final door position top after interruption of the photocell.

The photocell optic can be cleaned with a soft cloth and water upon requirement.

7.6 Manufacturer's certifications

7.6.1 Declaration of Incorporation

Albany Door Systems declares that the RapidProtect RP 2000 is an incomplete machine under the Machinery Directive 2006/42/EC. The product is a machine protection door with the property of separating the hazardous areas of another installation and thereby reducing the risk to people and goods.

Commissioning of our door is prohibited until it has been determined that the installation incorporating our door complies with the provisions of the EC Machinery Directive.

7.6.2 EC Declaration of Conformity

We, Albany Door Systems, declare that a door sold by us meets the fundamental health and safety requirements of all relevant European Directives and is therefore compliant with these Directives. The EC Declaration of Conformity is the basis for CE marking of a complete machine/installation.

7.6.3 Prototype test

The submission of a prototype test represents an additional voluntary proof on the part of the manufacturer over and above compliance with "Conformity". A state-run, state-commissioned or state-approved institution (e.g. professional association or TÜV) will, on request, test the conformity of e.g. a machine protection door based on the relevant current norms, regulations and standards specified for their intended purpose and certify this conformity by issuing a prototype test.

The prototype test remains valid until:

the period of validity stated in the prototype test elapses

or

there is a change to, or nullification of, the relevant norms, standards and regulations in respect of

issue of the prototype test

or

there is a change in the scope, design or type of use of the product that is the subject of the prototype test